Generative AI Versus Human Judgment in Supplier Selection:
An Expert-Benchmarked Behavioral Experiment in Final Supplier Selection

AI in Behavioral Supply Chain Management – Seminar Final Report (Research Proposal)

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1. Introduction and Research Question

Supplier selection is a high-stakes procurement decision: choosing the "wrong" supplier can propagate into higher total costs, quality failures, delivery disruptions, and reputational damage. While many organizations rely on multi-criteria scoring tools, real-world supplier evaluation rarely collapses into a purely mechanical ranking. Buyers must integrate quantitative KPIs (e.g., price, lead time, defect rates) with qualitative evidence (e.g., audit notes, relationship quality, communication reliability), while managing time pressure and stakeholder scrutiny (Taherdoost & Brard, 2019).

Generative AI (GenAI), especially large language models (LLMs), is increasingly positioned as a decision technology that can synthesize heterogeneous information and generate recommendations at scale. Recent practitioner-oriented work argues that GenAI can accelerate supply chain decision-making and improve managerial productivity by converting documents and dispersed information into actionable insights (Menache et al., 2025). Yet, organizational adoption depends not only on technical capabilities but also on behavioral responses: How decision makers and stakeholders perceive AI-driven decisions in terms of legitimacy, fairness, transparency, and trust. Evidence from behavioral operations and related domains shows that people may under-rely on algorithms after observing mistakes (algorithm aversion) (Dietvorst et al., 2015) but may also prefer algorithmic judgments in some contexts (algorithm appreciation) (Logg et al., 2019). In addition, LLMs can exhibit human-like biases and framing sensitivities in managerial decisions (Chen et al., 2025), raising questions about when delegating decisions to GenAI is beneficial.

In procurement practice, supplier choice often follows a two-step pipeline: (i) a specialist (or team) screens a large pool and produces a shortlist of qualified candidates, and (ii) a final decision-maker selects the winner from that shortlist. In this paper, we focus on step (ii), where information is more complete and comparable across candidates, making the decision amenable to structured analysis; yet still behaviorally non-trivial because decision makers must integrate quantitative KPIs with qualitative evidence and interpret risk and uncertainty. Moreover, "optimality" in supplier selection is contestable rather than mechanically determined by a single scoring formula; in practice, firms often justify high-stakes choices via committees and expert judgment. This setting therefore allows us to study delegating the final choice to GenAI as a behavioral problem: with identical dossiers, human and GenAI decision makers may differ in decision quality, efficiency, and perceived legitimacy.

Research question. Given a human-generated shortlist of qualified suppliers with complete, comparable information, does allowing a GenAI system to make the final supplier selection (relative to a human final decision-maker) improve decision outcomes and shape perceptions of legitimacy?

We operationalize outcomes along three dimensions: (1) decision quality relative to an expert benchmark, (2) efficiency (decision time and justification completeness), and (3) perceived legitimacy (trust, perceived fairness, and perceived transparency/understanding) at the time of the decision.

Our proposal contributes to the objective of analyzing AI adoption challenges in supply chain management from a behavioral lens: even if GenAI can process information systematically, organizations must manage reliance, acceptance, and accountability. We propose a controlled experiment that isolates the causal effect of delegating the final choice to GenAI, using a benchmark defined by a small panel of expert judges (faculty) and measuring both objective alignment with that benchmark and subjective legitimacy perceptions.

2. Literature Review

We structure the literature review around three streams that motivate the hypotheses and design.

2.1. GenAI as a decision technology in operations and supply chains. Recent literature highlights that AI systems can augment operational decision-making and reshape work processes (Yin et al., 2025). In bargaining and contracting settings, LLM agents can influence information processing and outcomes, making the interaction between AI outputs and human interpretation central (Kirshner et al., 2025). In a supply-chain context, these findings suggest that delegating supplier choice to GenAI may change both the quality of the final decision and the decision process (speed, consistency, explanation style).

2.2. Behavioral responses to algorithmic decisions: reliance, aversion, and appreciation. A foundational challenge for AI adoption is "appropriate reliance": users may either distrust useful recommendations or over-trust unreliable ones. Algorithm aversion shows that people may avoid algorithms after observing error, even when the algorithm is objectively better on average (Dietvorst et al., 2015). Conversely, algorithm appreciation documents contexts in which decision makers prefer algorithmic advice over human judgment (Logg et al., 2019). Recent operations research further distinguishes fast, intuitive reliance from slower, deliberative evaluation of algorithmic advice (Snyder et al., 2025). For supplier selection, the key implication is that performance gains from GenAI are not sufficient for adoption; perceived legitimacy and trust dynamics matter.

2.3. Transparency, fairness, and trust in AI-supported decisions. Beyond performance, stakeholders evaluate decision processes. LLM-based systems can provide narrative rationales, but these may be perceived as "black-box" or may crowd out unique human knowledge if users overweight AI explanations (Chen, Gao, & Liang, 2025). In algorithmic decision contexts such as hiring, perceived fairness can be higher for algorithmic decisions when they are seen as more consistent and less biased (Choung et al., 2023). For procurement, perceived procedural fairness and transparency are particularly salient because suppliers and internal stakeholders may contest decisions. Therefore, a procurement-relevant evaluation of GenAI should include both objective decision quality and subjective legitimacy, including perceived trust and transparency at the time of the decision.

2.4. Research gap. While prior work studies algorithmic advice and AI workflows, there is limited behavioral evidence on delegating the final supplier selection decision to GenAI under realistic procurement dossiers and an expert-defined benchmark of "best choice." In addition, much of the existing evidence focuses on advice-seeking settings (where humans can accept or ignore recommendations), whereas procurement governance increasingly involves formally delegating decisions and documenting rationales. By combining an expert benchmark with measures of efficiency, justification quality, and perceived legitimacy, the proposed study examines when GenAI delegation improves decision outcomes and when legitimacy concerns may constrain adoption.

3. Experimental Design

3.1. Overview and treatments. We propose a between-subject experiment with two conditions that use the same supplier dossiers and shortlist but differ in whether the final selection step is made by the participant (control) or delegated to GenAI (treatment).

- Control (Human → Human): Participants act as the final procurement decision-maker and select one supplier from a human-generated shortlist.
- Treatment (Human → GenAI): Participants observe that the organization delegates the final choice to a GenAI system (e.g., ChatGPT). The GenAI selects the supplier from the same shortlist and provides a brief written rationale.

This design isolates the behavioral impact of delegating final selection to GenAI, keeping shortlisting fixed and information constant.

3.2. Procedure and checks. Participants complete a short background questionnaire (e.g., procurement experience, numeracy, and prior GenAI use) and then evaluate 2–3 independent supplier-selection scenarios. In the Human → Human condition, participants select a supplier and provide a brief written justification. In the Human → GenAI condition, participants review the GenAI-selected supplier and rationale and indicate endorsement (e.g., accept the recommendation for the organization) before answering the same perception measures. After each scenario, we include light manipulation/attention checks (identify which supplier was selected and who made the final decision) and a brief recall item (name one KPI or qualitative note that most influenced the decision) to ensure engagement with the dossier and rationale. Decision time is recorded at the scenario level.

3.3. Task materials: procurement dossiers. Each participant evaluates 2–3 independent supplier-selection scenarios. Each scenario includes a shortlist of 4 finalist suppliers (e.g., shipping-box suppliers for a European consumer-goods firm) with:

- A structured KPI table (price, lead time, defect rate, ESG/compliance score, capacity, past on time delivery rate).
- Clear constraints (minimum ESG score, minimum capacity, compliance requirement).
- Short qualitative notes per supplier (e.g., audit comment, incident summary, communication reliability) to reflect realistic soft evidence.

To avoid a trivial "compute-and-pick-max" task while still keeping information complete, we frame supplier performance as inherently uncertain: the KPI table summarizes historical central tendencies and variability (e.g., mean and volatility of late deliveries), rather than deterministic future outcomes.

3.4. Expert benchmark and elicited decision logic. Decision quality is evaluated against an expert benchmark defined by a small panel of faculty judges (e.g., 2–3 professors/PhD-level instructors with operations/procurement expertise). The benchmark is constructed in three steps To keep expert workload feasible, we target 15–25 scenarios and use a standardized rating form; each scenario review is expected to take approximately 5–7 minutes.:

- Step 1 (Independent expert choices): Each expert reviews each scenario dossier and selects the best supplier. Experts also identify a "near-optimal set" (e.g., top-2 acceptable choices) when the scenario is genuinely ambiguous.
- Step 2 (Elicitation of expert evaluation logic): To document how experts decide, each expert completes a short, structured form that elicits (i) criterion weights (relative importance), (ii)

disqualifying constraints, and (iii) risk attitude toward delivery/quality uncertainty. This elicits an implicit utility model (e.g., a weighted additive utility with penalties for constraint violations and risk), not as the benchmark itself but to make the benchmark auditable and interpretable.

- Step 3 (Consensus benchmark): The benchmark winner is defined by majority vote; ties are resolved through a brief consensus discussion (Delphi-style) documented in writing. We report inter-rater reliability using both percent agreement and a chance-corrected statistic (e.g., Fleiss' kappa) and document cases labeled as genuinely ambiguous via the expert-defined near-optimal set. Decision quality is measured as (a) match with the benchmark winner and (b) whether the chosen supplier falls within the near-optimal set.

This approach aligns with real procurement practice, where "optimality" is normative and expert-driven rather than mechanically determined by a single scoring formula.

3.5. Measures.

- Primary outcome: Decision quality. (i) Indicator that the decision matches the expert benchmark winner; (ii) indicator that the decision falls within the near-optimal set; (iii) optional distance measure if experts provide an ordinal ranking.
- Secondary outcomes: (a) Decision time (from dossier exposure to final choice/endorsement), (b) justification quality, and (c) perceived legitimacy. Justification quality is coded using a transparent rubric capturing (i) coverage of stated criteria and constraints, (ii) evidence citation (explicit reference to relevant KPIs and qualitative notes), (iii) acknowledgement of uncertainty/trade-offs, and (iv) internal coherence. Perceived legitimacy is measured after each scenario using multi-item Likert scales capturing trust, perceived fairness, and perceived transparency/understanding.
- Controls and moderators: procurement experience, numeracy, and prior GenAI usage; exploratory analyses can test whether individual differences moderate AI acceptance.

3.5. GenAI specification (input/output standardization). In the GenAI condition, the model receives exactly the same dossier as participants, including the KPI table, constraints, and qualitative notes. The prompt instructs the model to (i) select one supplier, (ii) explicitly cite which KPIs/notes drove the decision, and (iii) acknowledge uncertainty (e.g., delivery variability) rather than presenting deterministic certainty. To avoid run-to-run variance, we pre-generate and fix the GenAI output per scenario using a standardized prompt and the same model configuration. Because the GenAI decision is fixed at the scenario level, comparisons of decision quality treat scenarios as the unit of observation for the GenAI

condition; accordingly, we construct a sufficiently large scenario set (e.g., 15–25 dossiers) and ensure each scenario is evaluated by multiple participants in the human condition.

4. Hypotheses and Planned Analyses

4.1. H1 (Decision quality): Relative to human final decision-makers, GenAI final selections will align more frequently with the expert benchmark winner (and/or fall more often within the expert-defined near-optimal set). The mechanism is systematic integration of structured KPIs and constraints, reducing noise from attention limits and inconsistent weighting.

4.2. H2 (Decision efficiency & justification): Delegating the final choice to GenAI will reduce decision time and produce more complete criterion-based justifications (higher coverage of stated KPIs/constraints) than human decisions. This follows from GenAI's ability to summarize multi-attribute evidence quickly and present a structured rationale.

4.3. H3 (Perceived legitimacy): Compared to human-made decisions, GenAI-made decisions will be perceived as less legitimate at the time of the decision (lower trust and lower perceived transparency/understanding), reflecting accountability concerns and "black-box" perceptions in high-stakes procurement. We expect perceived fairness to depend on whether participants view the process as consistent and impartial; therefore, fairness is measured as part of the legitimacy construct and analyzed separately as an exploratory component.

4.4. Planned analyses. Because the GenAI decision is fixed per scenario, H1 is assessed primarily at the scenario level. For each scenario, we compute (a) whether the GenAI choice matches the expert benchmark (and/or falls within the near-optimal set) and (b) the proportion of human participants (Human → Human) whose choices match the benchmark/near-optimal set. We compare GenAI match indicators to human match rates across scenarios using paired comparisons across scenarios and complementary mixed-effects models (with scenario random effects; participant random effects for human choices) and report cluster-robust standard errors. For H2, we analyze decision time and justification quality. Decision times are compared using mixed-effects models at the scenario–participant level, while justification quality is scored with the predefined rubric; GenAI justification is evaluated once per scenario and compared to the distribution of human justifications within the same scenario. For H3, we analyze legitimacy ratings after each scenario using mixed-effects models with scenario random effects, controlling for procurement experience and prior GenAI familiarity; fairness and transparency

components are examined separately as exploratory decompositions of the legitimacy construct. We report effect sizes and confidence intervals and conduct robustness checks using alternative benchmark definitions (e.g., majority winner versus near-optimal set). Target feasibility for inference is achieved by using a sufficiently large scenario set (e.g., 15–25 dossiers) and a participant sample that yields multiple human evaluations per scenario (e.g., approximately 100–140 participants completing 2–3 scenarios each).

4.5. Feasibility and ethics. The design requires only standard survey/experiment infrastructure and synthetic but realistic dossiers; no deception about real suppliers is necessary. The expert-benchmark step is feasible within a faculty time budget by limiting the scenario set (e.g., 15–25 dossiers) and using a structured rating form. Participants provide informed consent, and data are collected anonymously; if student samples are used, we transparently discuss limits to external validity and include experience measures to support interpretation.

5. Expected Contributions and Limitations

5.1. Expected contributions. Practically, the study informs procurement teams when delegating final supplier choice to GenAI is likely to improve decisions and when behavioral barriers (legitimacy, acceptance, and accountability concerns) may undermine adoption. The expert-benchmark approach mirrors real governance: organizations often justify procurement choices by appealing to expert panels or committees, and the proposed design makes that benchmark explicit and auditable. By jointly measuring decision quality, efficiency, and perceived legitimacy, the study clarifies trade-offs that matter for responsible GenAI deployment in procurement.

Theoretically, the study extends behavioral operations insights on algorithm reliance (Snyder et al., 2025) and AI workflow timing (Yin et al., 2025) to a procurement decision that combines structured KPIs and soft evidence. It also complements evidence that LLMs can behave like human decision-makers in biased ways (Chen et al., 2025) by testing a high-relevance managerial task with explicit constraints and accountability.

5.2. Limitations. First, an expert benchmark is normative and may vary across experts; we mitigate this by documenting expert decision logic, reporting agreement (e.g., Fleiss' kappa), and using a near-optimal set. Second, because the GenAI output is fixed per scenario, inference about GenAI decision quality depends on having a sufficiently large and diverse set of scenarios; we address this by constructing 15–25

dossiers and sampling across product categories and trade-off profiles. Third, external validity may be limited if participants are students; recruiting participants with procurement familiarity and measuring experience can strengthen interpretation. Fourth, in our experimental setup, the control condition requires participants to actively select a supplier, whereas the GenAI condition primarily asks participants to review and endorse a proposed selection. As a result, the treatment varies not only the identity of the decision-maker (human vs. GenAI), but also agency/ownership (self-made vs. externally proposed decision) and effort (full decision vs. evaluation task). Consequently, estimated effects, especially on time and perceived legitimacy, should be interpreted as the impact of delegating the decision to GenAI rather than a pure "AI vs. human decision-maker" effect. Future research can disentangle these components using a symmetric review design (human rationale vs. GenAI rationale) or a 2×2 design that varies decision-maker and participant role separately. Finally, GenAI performance may vary by model and prompt; we mitigate this through standardized prompts, fixed outputs per scenario, and transparent reporting of model version and prompting details.

5.3. Robustness and sensitivity. To ensure the study is not reducible to a simple weighted-sum exercise, scenarios will be constructed to include genuine trade-offs (e.g., a low-price supplier with higher delivery variability versus a higher-price supplier with strong reliability) and qualitative evidence that is informative but not perfectly decisive. We will pretest dossiers to verify that (i) experts exhibit some disagreement (otherwise the scenario is revised), (ii) human choices exhibit meaningful dispersion, and (iii) the GenAI output is stable under the standardized prompt. As internal validity checks, we re-estimate main results under alternative benchmark definitions (e.g., majority-winner benchmark versus near-optimal set, and leave-one-expert-out benchmarks) to confirm findings are not driven by a single expert or a knife-edge consensus rule. For the justification rubric, a subset of responses will be double-coded and inter-coder reliability will be reported before full coding.

6. References

1. Chen, Y., Kirshner, S. N., Ovchinnikov, A., Andiappan, M., & Jenkin, T. (2025). A Manager and an AI Walk into a Bar: Does ChatGPT Make Biased Decisions Like We Do? Manufacturing & Service Operations Management, 27(2), 354–368. https://doi.org/10.1287/msom.2023.0279.
2. Chen, Z., Gao, R., & Liang, Y. (2025). Revealing AI Reasoning Increases Trust but Crowds Out Unique Human Knowledge. Working paper (preprint).
3. Choung, H., Seberger, J., & David, P. (2024). When AI is Perceived to Be Fairer than a Human: Understanding Perceptions of Algorithmic Decisions in a Job Application Context. International Journal of Human-Computer Interaction, 40(22), 7451–7468. https://doi.org/10.1080/10447318.2023.2266244.
4. Dietvorst, B. J., Simmons, J. P., & Massey, C. (2015). Algorithm Aversion: People Erroneously Avoid Algorithms After Seeing Them Err. Journal of Experimental Psychology: General, 144(1), 114–126. https://doi.org/10.1037/xge0000033.
5. Kirshner, S. N., Pan, Y., Wu, J. X., & Gould, A. (2025). Talking Terms: Agent Information in LLM Supply Chain Bargaining. Decision Sciences. https://doi.org/10.1111/deci.70010.
6. Logg, J. M., Minson, J. A., & Moore, D. A. (2019). Algorithm Appreciation: People Prefer Algorithmic to Human Judgment. Organizational Behavior and Human Decision Processes, 151, 90–103. https://doi.org/10.1016/j.obhdp.2018.12.005.
7. Menache, I., Pathuri, J., Simchi-Levi, D., & Linton, T. (2025). How Generative AI Improves Supply Chain Management. Harvard Business Review, 103(1), 86–95 (January–February 2025).
8. Snyder, C., Keppler, S., & Leider, S. (2025). Algorithm Reliance: Fast and Slow. Management Science. https://doi.org/10.1287/mnsc.2023.01989.
9. Taherdoost, H., & Brard, A. (2019). Analyzing the Process of Supplier Selection Criteria and Methods. Procedia Manufacturing, 32, 1024–1034. https://doi.org/10.1016/j.promfg.2019.02.317.
10. Yin, J., Ngiam, K. Y., Tan, S. S.-L., & Teo, H. H. (2025). Designing AI-Based Work Processes: How the Timing of AI Advice Affects Diagnostic Decision Making. Management Science, 71(11), 9361–9383. https://doi.org/10.1287/mnsc.2022.01454.